DAEN 427/ISEN 427 Decision and Risk Analysis

Fall 2025

Time:	T-R 9:35–10:50	**Place:**	ETB 1013
Instructor:	Alexander Abuabara	**Email:**	abuabara@tamu.edu
		Zoom:	https://tamu.zoom.us/my/abuabara

Office Hours: MWF 11:30am–1pm or by appointment. I'm usually in my office ETB 4013 during office hours. If you need a different time, I'll do my best to accommodate, please include the course number when reaching out. I'm happy to discuss anything except material covered during an unexcused absence. If other students are already present, feel free to join the conversation. If you would like to discuss a private matter, please schedule an appointment.

Prerequisites: Grade of C or better in ISEN 310, DAEN 321, or STAT 212.

Cross Listing: ISEN 427/DAEN 427

Description: Overview of the state of the art in descriptive and prescriptive theories of decision making under uncertainty with emphasis on the ways in which human decisions depart from normative models of rationality; analytical foundations stemming from several disciplines, economics, psychology, management science; application in engineering systems will be considered.

Learning Objectives:

1. Understand Bayesian approach to perception and decisions under uncertainty.
2. Implement computational models for Bayesian inference.

Course Outline:

Lecture	Topic (*tentative*)
1	Graphical modeling. Perception as inference. Bayesian inference.
2	Uncertainty and confidence. Bayesian inference under measurement noise.
3	Predicting behaviors, estimation (MLE, MAP).
4	Cue combination, learning as inference.
	1st midterm exam (in class)
5	Computational aspects of Bayes inference (MCMC).
6	Discrimination and detection. Binary classification.
7	Bayesian linear regression.
8	Combining inference with utility.
	2nd midterm exam (in class)
9	Rational choice under risk.
10	Some paradoxes, prospect theory.
11	Expectation maximization algorithm.
12	Variational inference. Bayesian inference with neural networks.
	Project (in class)

Recommended Textbooks:

- *Bayesian Models of Perception and Action: An Introduction*. W. J. Ma, K. Kording, D. Goldreich (2022)
- *An Introduction to Bayesian Inference, Methods and Computation*. N. Heard (2021)
- *A Course in Behavioral Economics*. E. Angner (2021)
- *Bayesian Modeling and Computation in Python*. O. A. Martin, R. Kumar, J. Lao (2021)

Grading Policy:

Attendance/Assignments 10%; 2x Mid-term Exams 30% each; Final Project 30%

Grades will be calculated on the basis of total points earned. The thresholds for letter grades are: A ≥ 90%, B ≥ 80%, C ≥ 70%, D ≥ 60%, F < 60% If deemed necessary, the grade thresholds may be curved at the instructor's discretion.

You are expected to attend all class lectures except for university excused absences. The university rule regarding excused absences can be found at https://student-rules.tamu.edu/rule07.

Assignments are expected to be completed individually unless otherwise specified. Generally, assignments are to help you learn the information, they are not intended to be a practice run for the exams. Typically, an exam will examine the same concepts, just in different way than the assignments. If you're only able to apply concepts in specific situations (i.e. the assignments questions) then you likely don't actually understand the concepts, you just understand the mechanics of the questions being asked in the assignment.

Likewise, the final project is to be completed individually. While it serves as a culmination of the course material, it is not simply a repetition of prior assignments. Rather than testing your ability to follow familiar patterns, the project is designed to assess your deeper understanding and ability to apply various core concepts in new and varied contexts. If you can only approach problems that closely resemble those seen in class, it suggests a surface-level grasp of the material. True comprehension involves adapting your knowledge to novel situations, which the final project aims to evaluate.

More instructions for assignments, exams, and projects will be provided in class.

Late work will not be accepted except for approved university excused absences.

Any disagreements regarding a grade received on any graded material must be discussed within one week of the return of the graded material. No grade will be changed beyond this limit. If you request a regrade, the entire submission will be regarded (i.e. if we made a mistake in your favor, this will also be corrected).

Academic Integrity: Don't cheat! If your temptation is to not turn in work that isn't yours, don't do it. Come see me instead and ask questions, I'm here to help. Content is cumulative, so the earlier, the better. Modeling, coding, data analysis are skills that will open many doors for you: start learning now and set yourself up for success! Follows the Texas A&M Student Honor Code; all code notebooks and documents scanned with plagiarism detector.

It is acceptable for you to discuss any assignments except exams with other students prior to the due date. After discussing an assignment with someone else, do not write anything that you do not understand and always use your own language on the assignments. Copying another student's writing directly, even if it is a portion of the assignment, is considered cheating and an Aggie Honor Code violation. No leniency is given for academic dishonesty.

Academic Honesty: The Industrial & Systems Engineering Faculty uphold the Aggie Honor Code as an axiom of academic excellence. We consider its sincere observance to be essential for membership in our department and Texas A&M. We extend you the trust conferred to those who faithfully adhere to our honor code. Abuse of this trust is intolerable, thus we will report and assign an extreme penalty to those who do not stand with us in preserving the integrity symbolized by the Aggie Honor Code:

"An Aggie does not lie, cheat, or steal or tolerate those who do."

In this course, we will request the Aggie Honor Office to assign a grade of F for any violation of the Aggie Honor Code, no matter how minimal the violation may be. The Honor Council Rules and Procedures can be found at https://aggiehonor.tamu.edu

The Americans with Disabilities Act (ADA): A federal anti-discrimination statute that provides comprehensive civil rights protection for persons with disabilities. Among other things, this legislation requires that all students with disabilities be guaranteed a learning environment that provides for reasonable accommodation of their disabilities. If you believe you have a disability requiring an accommodation, please contact Disability Services, currently located in the Disability Services building at the Student Services at White Creek complex on west campus or call 979-845-1637. For additional information visit https://disability.tamu.edu

Statement on Diversity and Inclusion: It is important to us that all students feel accepted and comfortable in this class. If for any reason you feel disrespected, uncomfortable, or unsafe in our classroom please discuss your concerns with us via the medium you are most comfortable with. If you do not feel comfortable communicating directly to us, please reach out to Dr. Ntaimo (department head).

Additional Information: All information related to the course, including handouts, will be made available on https://canvas.tamu.edu. All communication related to the course will be sent electronically via Canvas as announcements or delivered to your @tamu.edu mailbox. You are expected to check your email regularly.